

Title: Big Data Analytics for E-Commerce Retail Performance

Executive Summary

This report provides an extensive **Big Data Analytics** evaluation for a Bangladesh-based e-commerce retailer in a competitive market. It runs its operation in online platforms such as other renowned online platforms like Daraz, Pickaboo, and Shajgoj in Bangladesh. The study focuses on transforming multi-table retail datasets, including customer_dim, item_dim, store_dim, trans_dim, time_dim, and fact_table into meaningful insights through SQL-based data preparation and Tableau for data visualization. Big data analytics plays an enormous role in improving operational efficiency, enhancing strategic decision-making, and strengthening customer understanding. The study shows the influence of big data on business operation and performance, evaluates dataset quality, data preparation procedures, identifies key metrics and relationships, and provides strategic insights through visualization techniques. It also recommends improved governance, automation, and forecasting for the coming periods to guide future organizational development.

Project Overview

This project applies big data analytics to a Bangladesh-based e-commerce retail dataset to support business decision-making through SQL-based data preparation and Tableau visualisation. The dataset consists of multiple relational tables, including customer, item, store, transaction, time, and fact tables, which together provide a structured view of sales activity, customer behaviour, product demand, payment methods, and geographic performance.

The project focuses on transforming raw retail data into useful business intelligence. SQL Server Management Studio was used to load, clean, validate, and transform the data before visual analysis. Key preparation tasks included checking missing values, identifying duplicate records, standardising location names, and adding a country field to improve geographic mapping in Tableau.

After data preparation, Tableau was used to develop visual insights and dashboards covering top-selling products, customer distribution by telecom operator, sales performance by location, payment method preferences, sales trends, and forecasting. These insights were then used to generate business recommendations related to inventory planning, targeted marketing, digital payment promotion, location-based strategy, product bundling, and future sales planning.

The overall aim of the project is to demonstrate how structured e-commerce data can be transformed into actionable insights that improve operational efficiency, customer understanding, and strategic decision-making.

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1 Introduction:

The current era is a data-driven era. Data-driven decision-making has become the standard in modern business performance, particularly in customer-centric industries such as retail and financial services. Organizations or businesses increasingly depend on high-quality customer data, which they can use to manage relationships, comply with regulations, design targeted marketing, and reduce the risk of fraud. However, poor data quality can severely affect the value of analytics.

In Bangladesh, the e-commerce sector has experienced rapid growth as consumers' transition to digital platforms. This was increased after the coronavirus pandemic Corona virus in 2020 throughout the world, and by improved internet accessibility, diversified payment systems, and expanding logistic infrastructure.

2 Data Brief:

The report represents data analytics, and a visualization study using a sales dataset from <https://www.kaggle.com/datasets/mmohaiminulislam/ecommerce-data-analysis>. This dataset Contains:

- 1,000,000+ sales records
- 9,191 customers
- 726 stores
- 264 items

And many attributes in 6 tables in csv (comma separated values) form. The data appears to represent individual customers in Bangladeshi context.

2.1 Data Description:

Data can broadly be categorized as structured, semi-structured, or unstructured (Chen, Chiang, & Storey, 2012). The data that is complete, easy to use, or well organized is structured data. The data in the format of images, text, audio, emails, and hard to find its meaning is unstructured data.

The dataset used in the report is fully structured:

For example, In customer_dim table

- Each row represents a unique customer record
- Each column is a clearly defined attribute:
 - Customer_key: a customer identifier
 - Name: full name of the customer
 - Contact_no: 13 digit mobile number with the country code

- Nid: 13 digit national identification number.

All four variables are suitable for storage in Microsoft SQL Server Management Studio as a relational table. While this report analyzes only structured data, the organization could enrich this dataset with unstructured data sources (e.g. complaint texts or feedback emails) in the future.

3 Tools Used:

This project used a combination of database management, business intelligence, and data visualisation tools to clean, transform, analyse, and present insights from the e-commerce retail dataset.

3.1 Microsoft SQL Server Management Studio

Microsoft SQL Server Management Studio was used for importing, storing, cleaning, and transforming the six relational CSV tables: customer_dim, item_dim, store_dim, trans_dim, time_dim, and fact_table. SQL queries were applied to check missing values, identify duplicate records, standardise text fields, add a country column for geographic mapping, and prepare the data for further analysis.

3.2 SQL

SQL was used as the main data preparation and analysis language. It helped to inspect data quality, join related tables, create business queries, and extract meaningful information about product sales, customer groups, payment methods, location performance, and sales trends.

3.3 Tableau

Tableau was used to create interactive visualisations and dashboards. It supported the analysis of top-selling products, customer distribution by telecom operator, sales by location, payment method comparison, sales trends over time, and forecasting. Tableau was also used for geographic mapping after the country and location fields were cleaned in SQL Server.

3.4 Microsoft Excel

Microsoft Excel was used for initial data inspection, reviewing CSV files, checking column structure, and supporting early understanding of the dataset before importing it into SQL Server.

3.5 Kaggle Dataset

The project used a public e-commerce dataset from Kaggle. The dataset contains structured retail data with customer, product, store, transaction, time, and fact tables, making it suitable for relational database analysis and business intelligence reporting.

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4 Data Preparation and Cleaning:

According to recent systematic analysis, SQL remains the basic in business intelligence ecosystems due to its ability to standardize data structure, facilitate query and improve accuracy

in dashboards (Akte & Mamun, 2023). It is very important to preprocess Data for the better output by running the work on SQL Management Studio. For the queries of SQL to handle any disparity in relational databases efficiently. It will help to develop

4.1 Loading and Structuring the Data:

First, a database has been created and then the 6 CSV (Comma Separated Values) files have been imported. The CSV files are imported into Microsoft SQL Server using the Import Data Tier Application wizard and stored in the database.

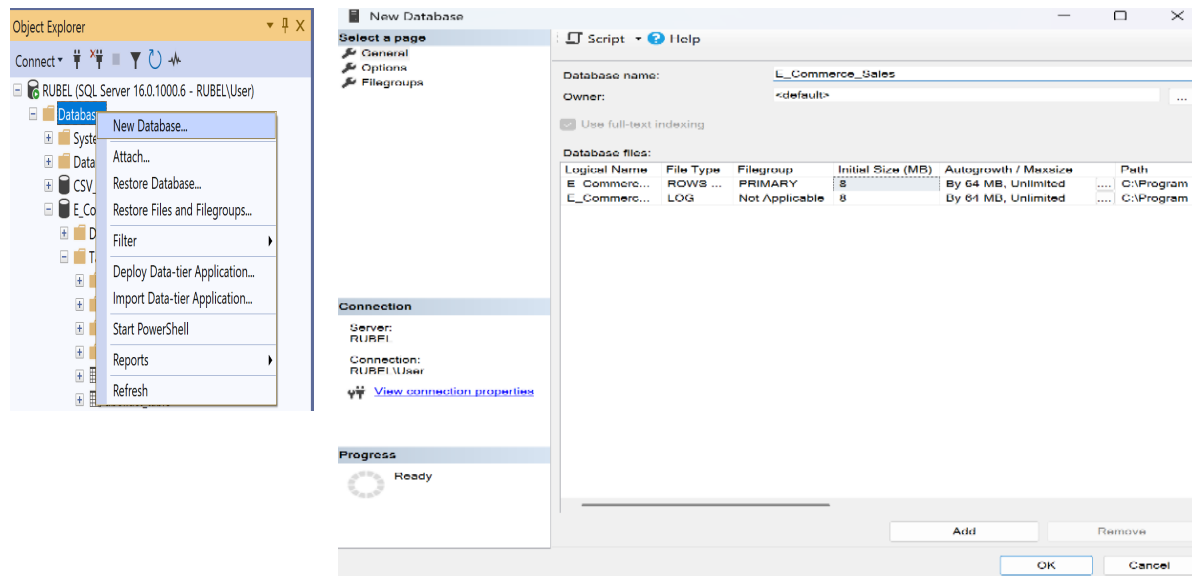


Figure 1 Create New Database

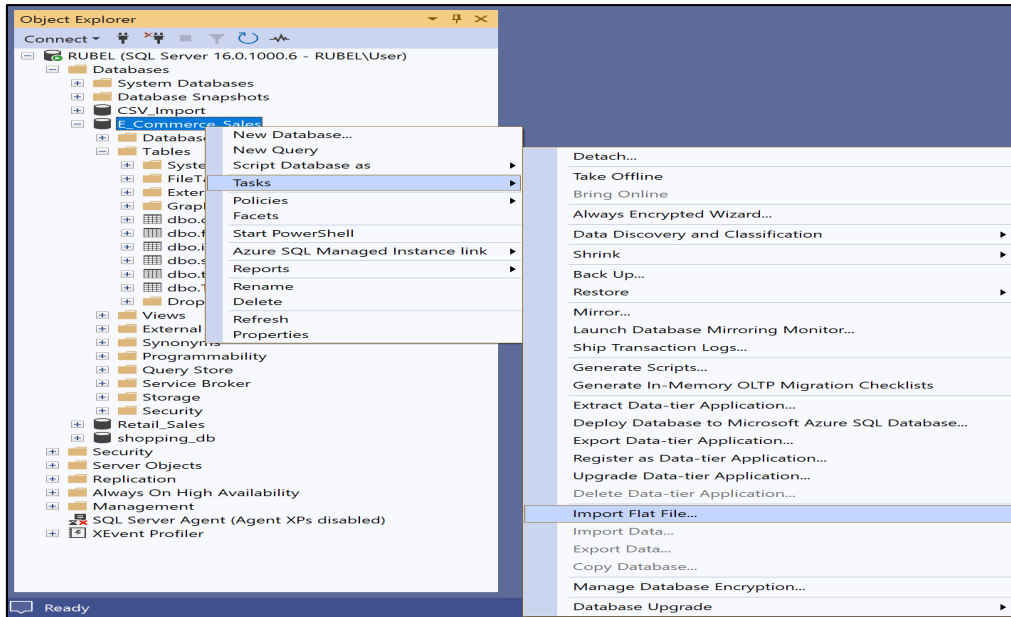


Figure 2 Import Files to Database

4.2 Handling Missing Values:

High quality data is must for accurate analytics, and the dataset is thoroughly inspected for missing values. The ‘name’ column contained several NULL values. Since removing those rows could lead to loss of data, a data cleaning strategy is used. NULL values are replaced with the label ‘Unknown’ using the SQL function.

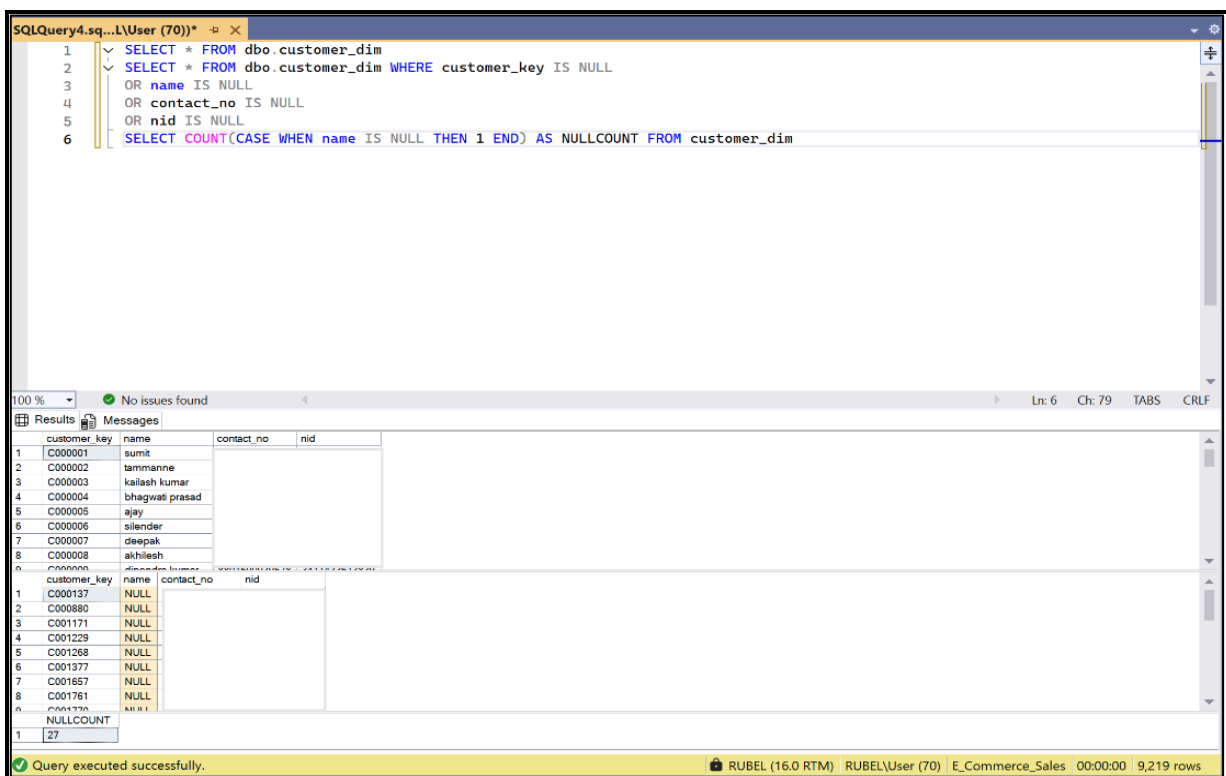


Figure 3 Handling Missing Values

4.3 Finding and Removing Duplicate Records:

During the data quality assessment stage, the dataset is examined for duplicate records, particularly within the key columns such as customer_key. Duplicate values can distort analytical result, inflate counts and shows the inaccurate insights. Therefore, a structured query was applied to identify and found no duplicate values for 'customer_key', and only 1 'contact_no' is found duplicate that is negligible.

The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows a query window with the following SQL code:

```

1 SELECT * FROM dbo.customer_dim
2 SELECT customer_key, COUNT(*) AS count_occurrence
3 FROM customer_dim
4 GROUP BY customer_key
5 HAVING COUNT(*) > 1

```

The bottom pane shows the 'Results' tab with a table of customer records. The table has four columns: customer_key, name, contact_no, and nid. The data is as follows:

customer_key	name	contact_no	nid
C000001	sumit		
C000002	tammanne		
C000003	kailash kumar		
C000004	bhagwat prasad		
C000005	ajay		
C000006	silender		
C000007	deepak		
C000008	akhilesh		
C000009	dipendra kumar		
C000010	nitin		
C000011	doodhnath pandit		
C000012	aslam allam		
C000013	rahul		

The status bar at the bottom indicates: 'Query executed successfully. RUBEL (16.0 RTM) RUBEL\User (61) E_Commerce_Sales 00:00:00 9,191 rows'.

Figure 4 Extracting Duplicate Records

4.4 Data Transformation:

Data transformation is important but often neglected stage in many data science pipelines. Applying the right transformation can enormously develop the performance of the learning algorithm. During the data preparation, some data transformation techniques were applied to improve the quality and the analytical value of the dataset. These transformations can ensure the data structure for SQL queries and Tableau visualization.

4.4.1 New field Addition:

During the data transformation stage, a new field name was added to 'store_dim' table. This step is essential for enabling accurate geographic mapping in Tableau. Geographic system requires a complete location hierarchy,

Country → Division → District → Upazila

Without a country field, Tableau is unable to correctly interpret Bangladesh by its longitude and latitude, and then its division that are consisted of several districts. District names (e.g., “Dinajpur”, “Cox’s Bazar”, “Comilla”, “Jessore”) may also be exist in other countries or do not uniquely map in the global geocoding system. As a result, Tableau produces null map points or locates them incorrectly on the map.

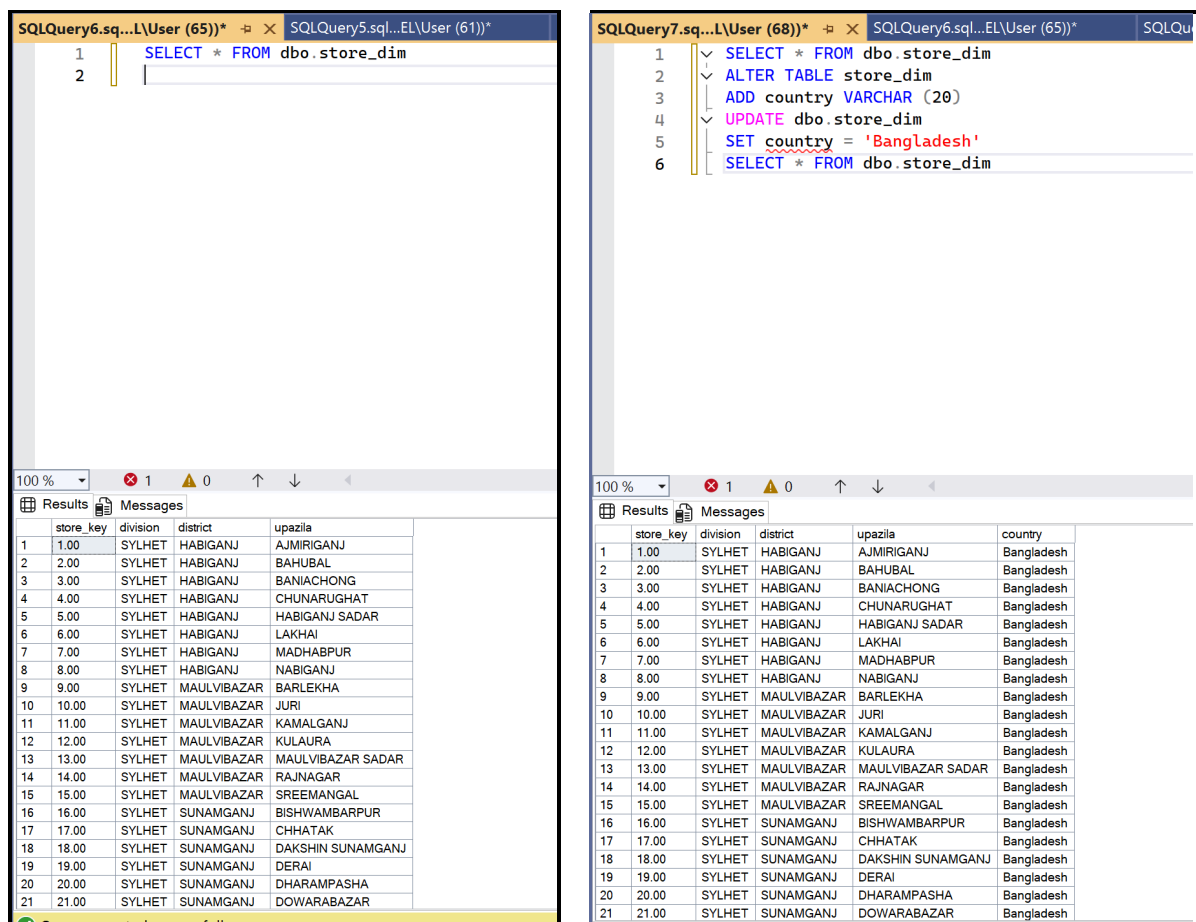
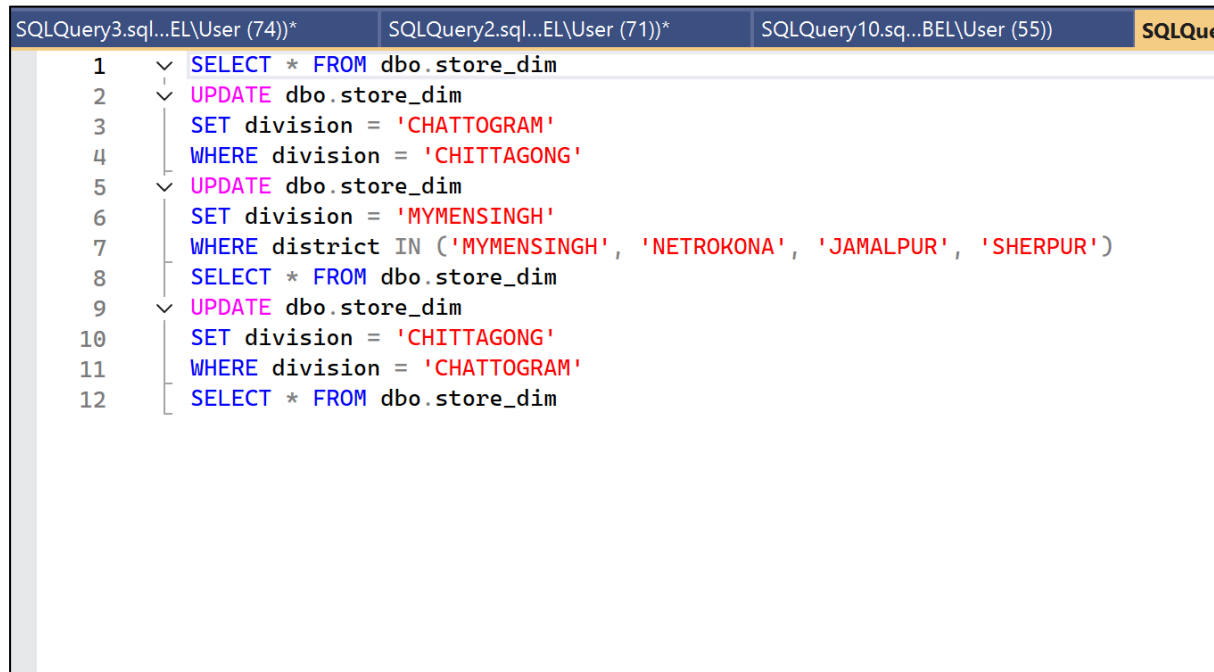


Figure 5 New Field Addition for Visualization of Map

By adding the name of the country, the map by Tableau can easily identify the geographical location of the division and districts of the country. This will help to predict the future growth by its location.

4.4.2 Text Standardization:

Text standardization was applied to ensure uniformity across all categorical and geographic fields. This included fixing spelling variations. These transformations significantly improved data integrity and helps to visualize it by Tableau.



The screenshot shows a SQL query editor with four tabs. The active tab is 'SQLQuery3.sql...EL\User (74))*'. The query is as follows:

```

1  SELECT * FROM dbo.store_dim
2  UPDATE dbo.store_dim
3  SET division = 'CHATTOGRAM'
4  WHERE division = 'CHITTAGONG'
5  UPDATE dbo.store_dim
6  SET division = 'MYMENSINGH'
7  WHERE district IN ('MYMENSINGH', 'NETROKONA', 'JAMALPUR', 'SHERPUR')
8  SELECT * FROM dbo.store_dim
9  UPDATE dbo.store_dim
10 SET division = 'CHITTAGONG'
11 WHERE division = 'CHATTOGRAM'
12 SELECT * FROM dbo.store_dim

```

Figure 6 Text Standardization

Here, the name of a division was corrected and set new division that was divided from Dhaka division very recently to use the correct geographical location on the map.

5 SQL-Based Analysis:

The dataset used in the report represents retail sales, consisting of six interrelated tables:

Table	Description	Key Field
customer_dim	Customer details (name, contact number, national ID)	customer_key
fact_table	Transactional data including quantity, price, total price, and foreign keys to dimension tables	payment_key
item_dim	Product information such as item name, description, supplier, and manufacturing country	item_key
store_dim	Store location data including division, district, and upazila	store_key
time_dim	Temporal information	time_key
trans_dim	Transaction method details	payment_key

5.1 Database Diagram:

Database Diagrams are extremely important in SQL and data analytics as they indicate the structure, relationships, and logic of database visually. A diagram helps to see the data flow and relationships among the tables. It also helps maintain data integrity, and identify errors and redundancies.

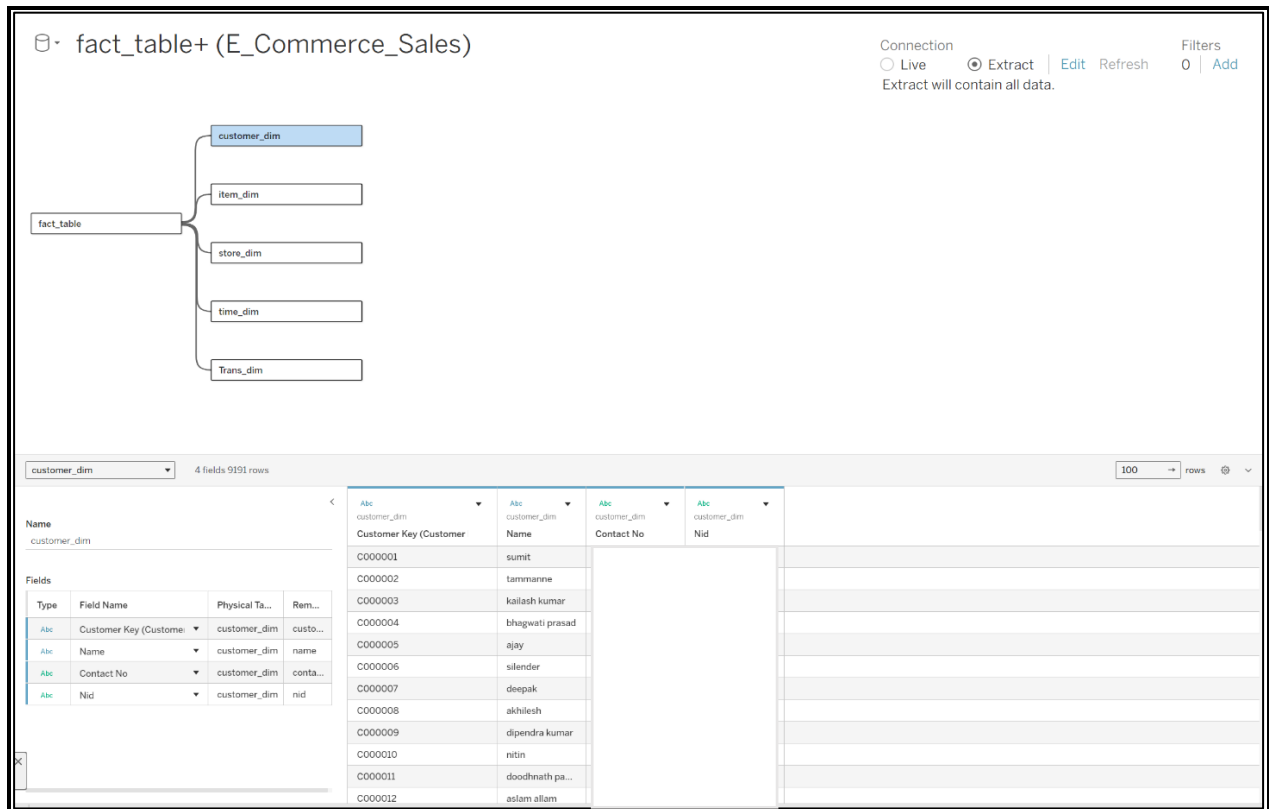


Figure 7 Making Relationships among Tables

There exists many to many relationships among these tables. Fact_table is in the center and other tables are connected to this table.

5.2 Strategic Queries:

From the relations of the tables, there arise some queries for strategic insights for the betterment of the organization and future prospects. The queries are:

5.2.1 What are the top selling products?

By this query, the organization can understand that which products will generate more revenue.

5.2.2 How are the customers distributed across telecom operators, and which operator group generates the highest revenue?

It will make understand that some special event or some special benefit will be given to those operator groups for revenue growth.

5.2.3 Which division (location) generate the highest total sales?

It will help the organization to understand where the supply of the products will have been increased, and the warehouse will be established in greater number.

5.2.4 Which payment methods are more frequently used?

It will help to identify the customers' payment preference, and show which bank has the highest number of customers.

5.2.5 What is the trend of total sales over a period of time?

It will make clear that how will be the coming days for the organization. Where the concentration will be given, and how to generate growth.

6 Tableau Dashboard Insights:

For the data visualization and analytics, Tableau is used in the report. It is one of the most useful tools for showing the insights of a dataset with its own query language only the drag and drop method by a mouse. The users can get quick results from the data and make predictions for the future. Through visual analysis with Tableau, patterns and relationships are discovered that might not have been known to exist.

6.1 Data Preparation for Tableau:

Before starting to work with Tableau, the data type must be checked for its correct format. There are some issues in the data that need to be fixed. The data should be shaped so that Tableau can interpret it. Data type should be corrected to string, varchar, number, or date as its type of data by which Tableau can calculate or not for processing results.

Furthermore, Tableau had to be connected to SQL Server so that it could show the

6.2 Insights Analysis:

Business Intelligence tools contribute to show transparent, real-time financial pictures, which can improve budgeting, investing, and performance accountability, (Zamil et al., 2024)

6.2.1 Insight 1: Top-Selling Items of the Organization

Using a Tableau bar chart, the highest-selling item was marked by total price. This visualization revealed the reflection of customer behavior for the product.

- The top-selling items generate the highest revenue due to a combination of demand and consistent purchase frequency.
- Strategic promotions will be given for best-selling items.
- Visibility of these items should be improved for the customers.

Business Insights:

- The retailer should ensure continuous stock availability of the top-selling items.

- Poor-performing items should undergo price adjustments, and make them as bundled with the top-selling products.

This metric is essential for **inventory management**, **profit optimization**, and **supply chain planning**.

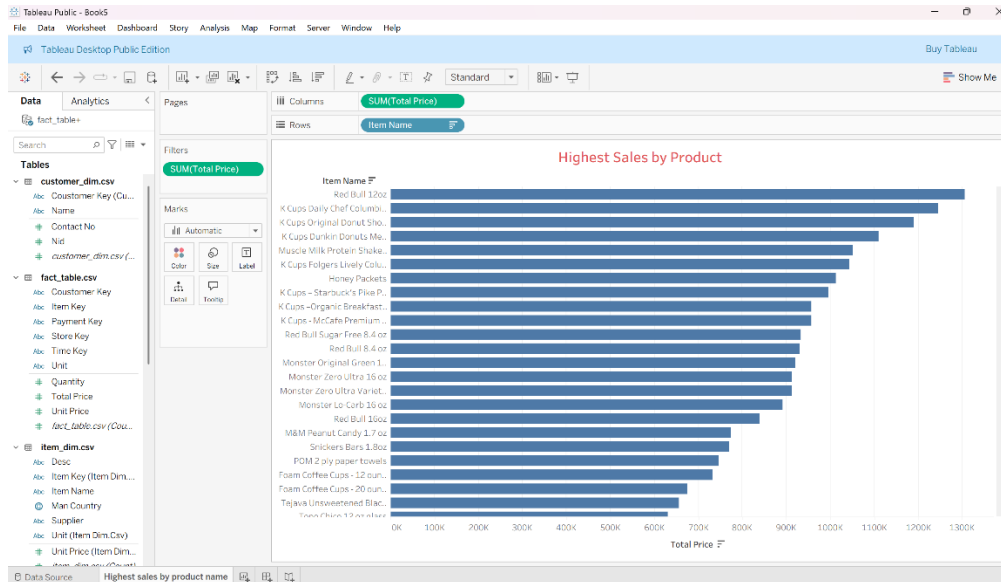


Figure 8 Top-selling items

6.2.2 Insight 2: Number of Customers by Telecom Operators

The customer contribution across mobile phone operators reveals a clear dominance, which accounts for the highest number of registered customers from Airtel followed by Banglalink, Teletalk, Grameenphone, and Robi.

- This pattern mirrors the potential opportunities for targeted and operator-based promotion strategies.



Figure 9 Number of Customers by Telecom Operator

6.2.3 Insight 3: Generate Highest Sales by Location

Geographical location is much more important consuming a product by its demands. How many people are living here and the number of customers are purchasing the targeted product. Visual analytics applied to market-level data enables identification of customer trends, products gaps, and growing opportunities that support market expansion strategies. (Thakur & Das, 2023)

- **Dhaka** division generates the highest sales due to its dense population and is the capital city of the country. The people living here have a higher purchasing capacity
- As the central economic hub of Bangladesh, Dhaka shows the urban consuming patterns, and increased retail activity.
- This insight highlights targeted operational or marketing strategies.

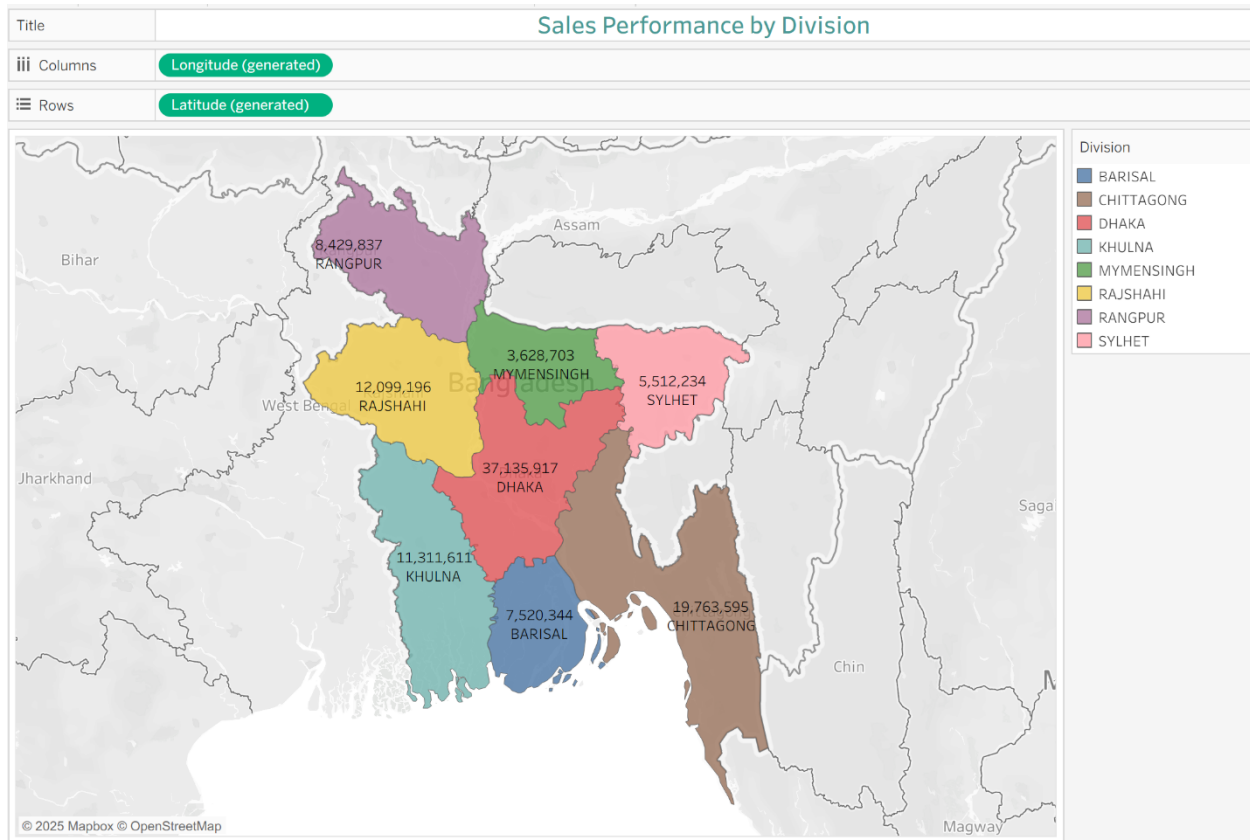


Figure 10 Highest Sales by Location

By this map, it is clear that, a greater portion of revenue comes from the capital city of the country 'Dhaka'. The organization will give emphasis to supply the products in this division continuously.

6.2.4 Insight 4: Analysis of Payment Method of Customer

- Pie Chart below shows the payment preferences of the customer
- Card payment method stands in the first position followed by Mobile banking wallet and Cash.
- The organization may give promotional points or cash back for the customers who use digital payment system to increase sales.

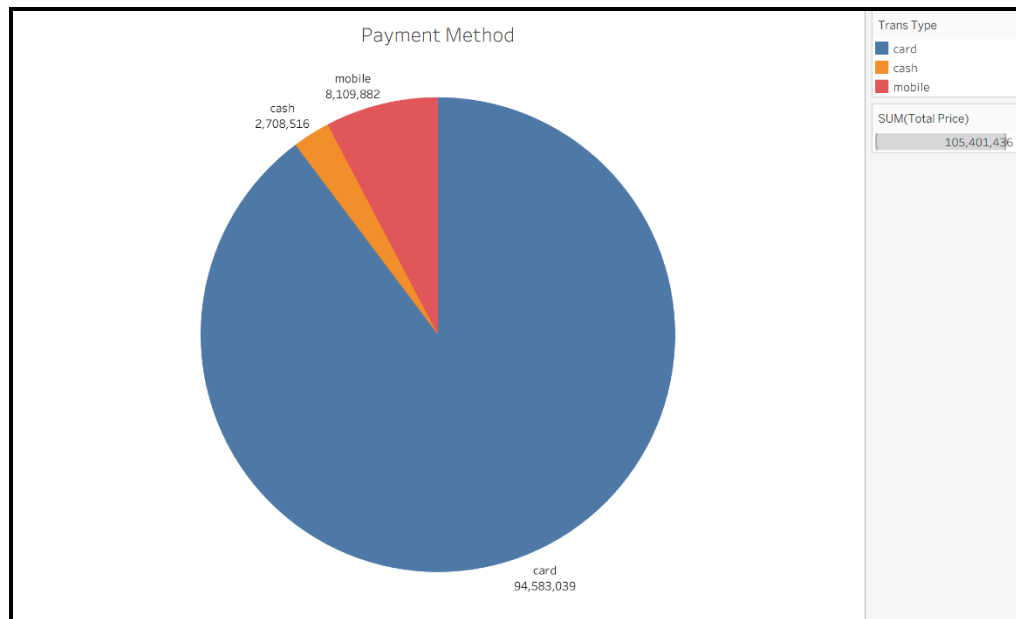


Figure 11 Payment Method Comparison by Customer

6.2.5 Insight 5: Trend of Sales Over Time:

- The line chart shows the analysis of sales pattern over the period of time.
- The time series analysis of monthly sales shows a fluctuating pattern throughout the year, indicating seasonal variation in customer demand.
- By this insight, the organization can give emphasis on inventory planning and marketing strategies with the observed seasonal pattern.

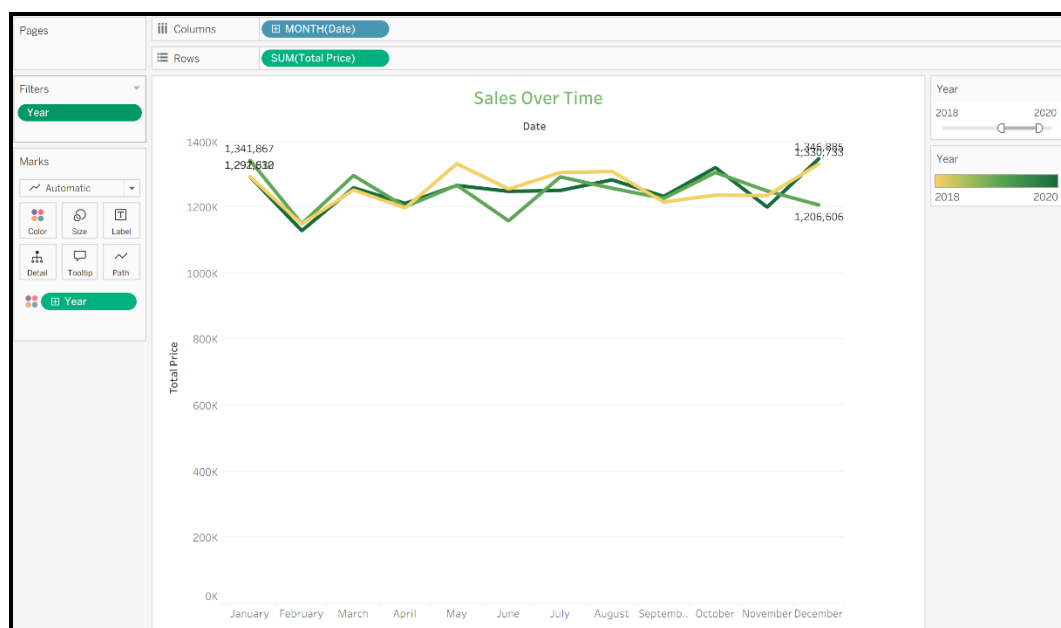


Figure 12 Sales Trend Over Time

6.3 Dashboard Design:

Real-time dashboards developed through Tableau enhance organizational responsiveness by providing timely access to financial, operational, and customer-level metrics, thereby supporting quick strategic decisions (Frempong et. al., 2022).

Multiple dashboards were developed to provide a multi-perspective analysis of sales performance.

- Sales Overview Dashboard: visualizing total revenue by month, year, and location.
- Customer Analysis Dashboard: profiling customer segments by purchase frequency and total spend.
- Product Performance Dashboard: ranking top-selling items and suppliers.
- Payment Method Dashboard: analyzing revenue distribution across cash, card, and bank transactions.

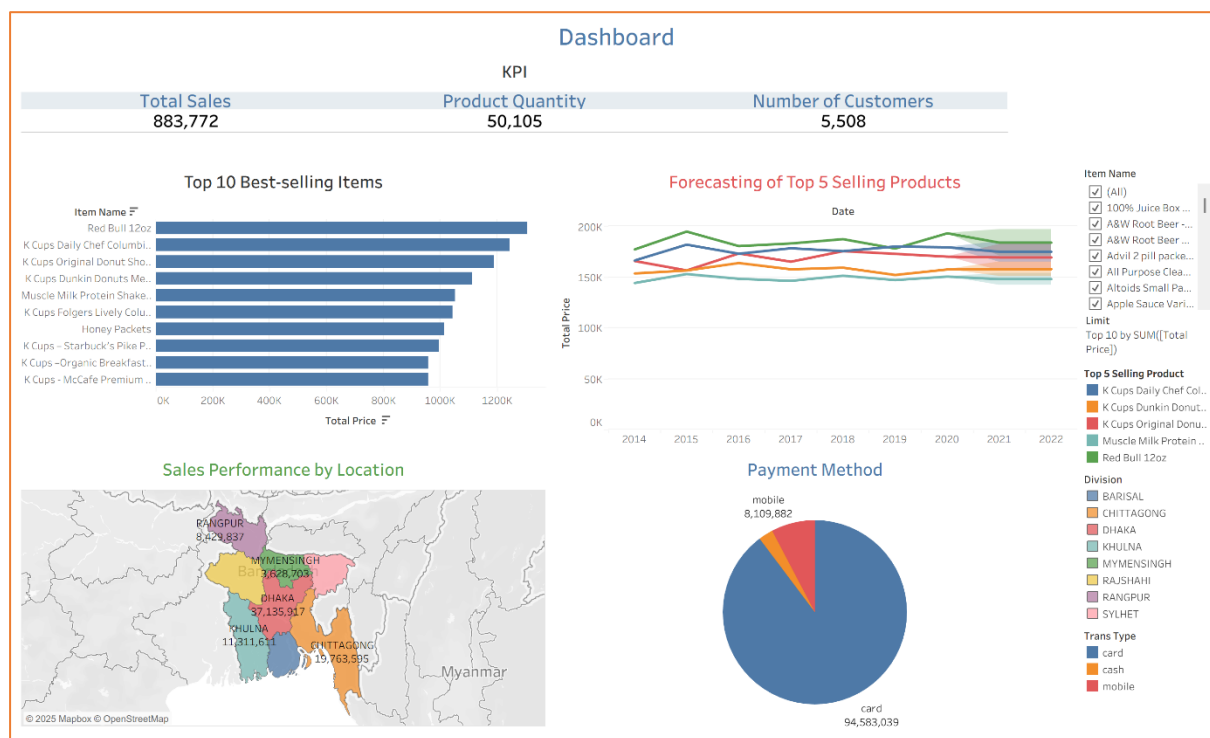


Figure 13 Designing Dashboard

7 Forecasting Analysis:

The forecasting model was developed by using historical yearly sales from 2014 to 2021. The historical series shows moderate year to year fluctuations but maintains a general consistent downward for the next two years. But it is not the trend or seasonality.

Forecasting of Total Sales by Location:

If we look on the forecasting for the next 2 years of sales by location (division), we found the revenue growth in all the locations but not over 5%. The organization give emphasis on its strategy of sales and marketing.

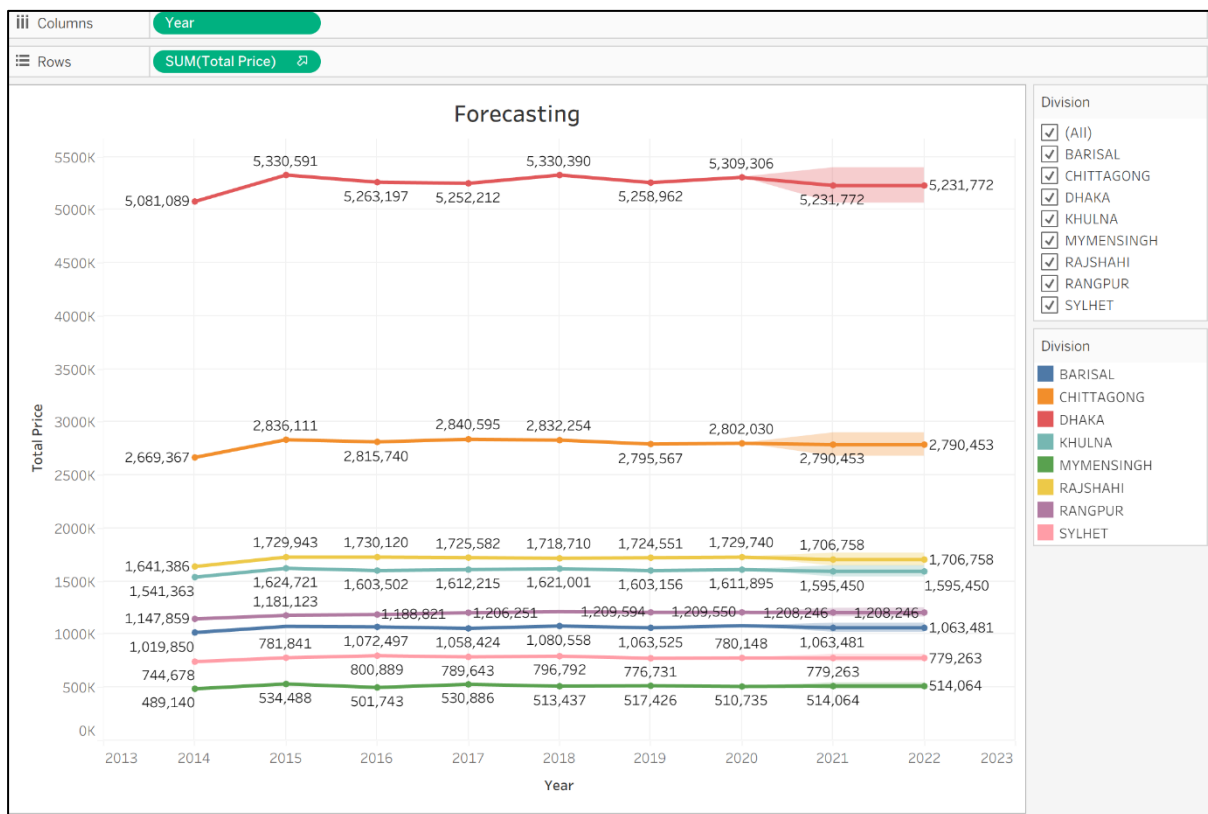


Figure 14 Forecasting by Location

Forecasting of Total Sales:

Forecasting by total sales for the next 2 years is not satisfactory. There will make a significant change in the strategy in the supply chain, and marketing policy, clustering of products, bundling items. The decision making authority would take prompt measures.

- It should apply loyalty programmes such as cashback offer, membership points, buy 2 get 1 for selected items etc.
- Should optimize pricing strategy through dynamic adjustments rather than flat discounts.
- Should increase stock availability.

- Should arrange training to the sales teams as well as marketing team.

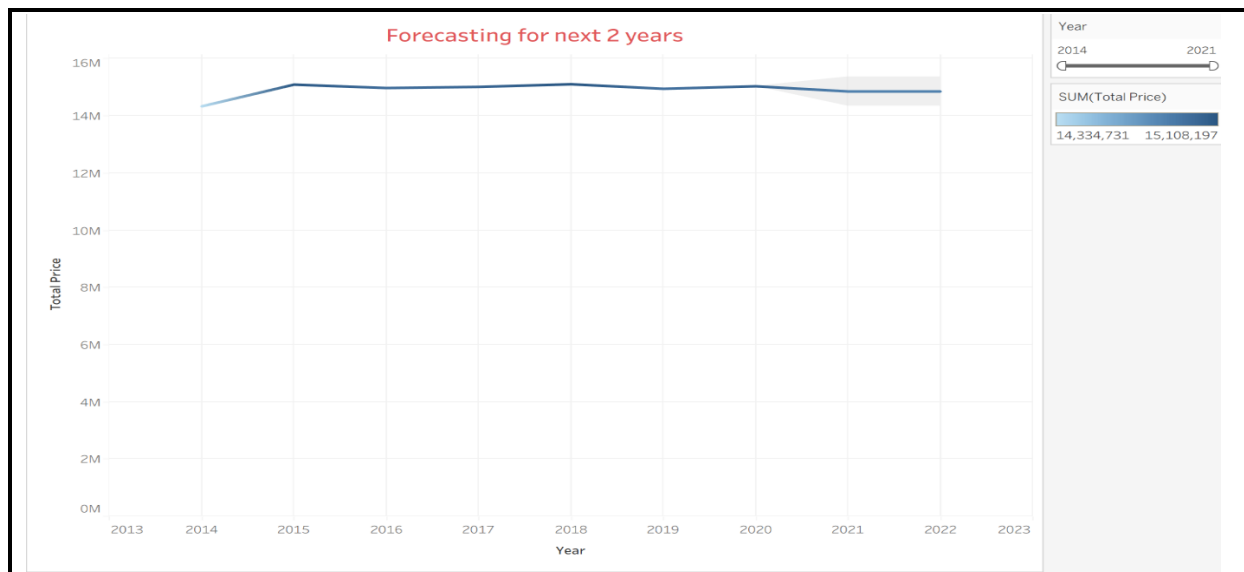


Figure 15 Forecasting of sales



Figure 16 Forecasting Options

- Maintain strategic relationships with supplier to expand geographical sales.
- Strengthen digital marketing through different social media.

8 Business Recommendations:

This report illustrates the significant influence on Big Data analysis of Big Data for growing business strategy, practice, and the performance for the selected Bangladesh-based e-commerce retailer. Through the backend work in SQL Management Studio for data preparation and visualization techniques by Tableau, different powerful analyses and multi-dimensional insights were generated. These analyses will support the organization in many ways to make important decisions to grow the revenue for the coming days. By examining the data of that organization, some recommendations will be given to develop the strategy to increase revenue.

- **Clustering high value customer segment by their purchasing behavior, frequency, product preference.** It will increase cross-selling and upselling.
- **Develop dynamic pricing and promotional Intelligence.**
- There have opportunities to **spread the range of product as their market demand** for a better customer support.
- **Develop the marketing strategy** to improve customer responsiveness.
- Create **bundling of the poor-selling** product with the highest sales to increase sales.
- **Business intelligence training can be provided** to the employees who work with the sales analysis.
- **Develop user interface of the website** so that the customer can easily buy by clicking the mouse and get better shopping experiences.
- **Some campaigns will be arranged by the most revenue generated customer based on the mobile phone operator.**

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